

AMITH LAL

AI & Machine Learning Engineer | Generative AI, RAG Systems & LLM Applications | Python · Django

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PROFESSIONAL SUMMARY

Generative AI and Machine Learning Engineer with strong **Python** skills, focused on clean, secure, and maintainable code. Hands-on experience fine-tuning LLMs (**LoRA/QLoRA**), designing and testing prompts, and building **RAG-based systems** end to end — chunking, embedding generation, vector search, and evaluation (**RAGAS**). Experienced in creating and maintaining **API endpoints** and system architecture with a full Gen AI stack (local LLMs, vector DBs, **LangChain**, **LangGraph**, **Ollama**). **AWS** certified, with solid **SQL** and **ML** knowledge, plus **Git**, **CI/CD**, code reviews, **Docker**, and **Kubernetes**.

CORE COMPETENCIES

Gen AI Stack	Local LLMs (Qwen, Llama) · Ollama · Vector DBs (Qdrant, ChromaDB) · LangChain · n8n · RAG · Chunking · Embeddings · Prompt Design & Testing · Agentic Workflows · Claude Code · MCP Servers
Machine Learning	Scikit-learn · Predictive Modeling · Model Deployment · Data Pipelines
Languages	Python (strong) · SQL · Java
Backend & APIs	Django · Django REST Framework · FastAPI · RESTful APIs · API Design & Maintenance · Streamlit · System Architecture · Firebase
DevOps & Cloud	AWS (EC2, S3, IAM) – AWS Certified · Vercel · Docker · Kubernetes · CI/CD · Git / GitHub · Code Reviews
Databases & Storage	Supabase (PostgreSQL) · MySQL · SQLite · Firestore
Engineering Practices	Clean, Secure & Maintainable Code · OOP · Data Structures & Algorithms · API Integration · Postman

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineer

Jul 2024 – Present

Linx AS, Kochi, India

Gen AI Document Processing & Extraction Platform | Python, Django, Ollama, LLMs, RAG

- Developed Django-based **Generative AI** applications that leverage **LangChain**, **LangGraph**, and **RAG** to orchestrate **local LLMs (Qwen, Llama)** for ingesting, analyzing, and extracting structured data from diverse documents.
- Implemented **RAG-based pipelines** end to end — document chunking, embedding generation, and vector search (**Qdrant**, **ChromaDB**) — enabling accurate Q&A, AI summaries, and targeted data extraction from uploaded files; evaluated retrieval and generation quality using **RAGAS** (context precision, faithfulness, answer relevancy).
- Designed extraction outputs using **structured output formatting** (schema-constrained JSON) to ensure consistent, machine-readable results suitable for direct integration with downstream systems.
- Fine-tuned LLM models using **LoRA** and **QLoRA** techniques for improved domain accuracy and reduced inference costs, alongside orchestrating **AI agents with tool-calling workflows** via **LangChain**, **Ollama**, and **LangGraph**.
- Configured and integrated **MCP (Model Context Protocol) servers** to extend RAG system capabilities, enabling structured tool access for LLM agents; designed and iteratively **tested LLM prompts** to improve extraction accuracy and response reliability.
- Created and maintained robust **RESTful API endpoints** utilizing **Django REST Framework** and **FastAPI** for file uploads and extraction, owning system architecture and integrating directly with **SAP systems**.
- Used **Claude Code** for AI-assisted development; wrote clean, secure, and maintainable code with **Git** best practices and **code reviews**.

Albert Invent Data Migration & Utility Development | Python, Albert SDK, Pandas, Streamlit

- Executed complex data migration initiatives transferring critical R&D data from client legacy systems into the Albert Invent platform.
- Developed custom utility tools with Python and the Albert SDK to automate data extraction, validation, and loading; engineered efficient bulk-processing scripts ensuring high data integrity and minimal downtime.
- Built a **Streamlit**-based internal application to streamline and monitor the data migration workflow, providing a UI for validation checks and migration status tracking.

Cloud Engineering Intern

Feb 2024 – May 2024

Trainso Training Solutions, Bengaluru, India

- Worked on AWS cloud infrastructure operations (IAM, EC2, S3, security configurations); assisted in deploying containerized (Docker) applications and managing access control policies.

EDUCATION

B.Tech – Computer Science · APJ Abdul Kalam Technological University · CGPA: 9.34

2019 – 2023

CERTIFICATIONS & PUBLICATIONS

ML Specialization, Andrew Ng (2024) · IBM Applied AI Professional Certificate (2023) · Python for Data Science & AI, IBM (2023) · AWS Solutions Architect Associate (2023) · Publication: Personalizable Super-App Framework, NCASCD (2023)